File Server Role Theory

Server: SERVER01 (domain lab.com) | Client: client01 (Windows)

Task 1: On your Windows Server VM, create a shared folder named 'MusicLibrary' and set the share permissions so only the 'Students' group has read access.

A shared folder named MusicLibrary was created and its share permission was limited to Read for the Students group.

1. In Active Directory Users and Computers, created a security group named Students (inside the Students OU) and added the users Student and test user as members.
2. Created the folder C:\Shares\MusicLibrary, and copied the music files into it.
3. Opened Properties > Sharing > Advanced Sharing, ticked Share this folder, and kept the share name MusicLibrary (network path \\SERVER01\MusicLibrary).
4. Opened Permissions, removed Everyone, added Students and ticked Read (Allow) only.

Result: at the share level, only members of the Students group can reach the folder, and they can only read.

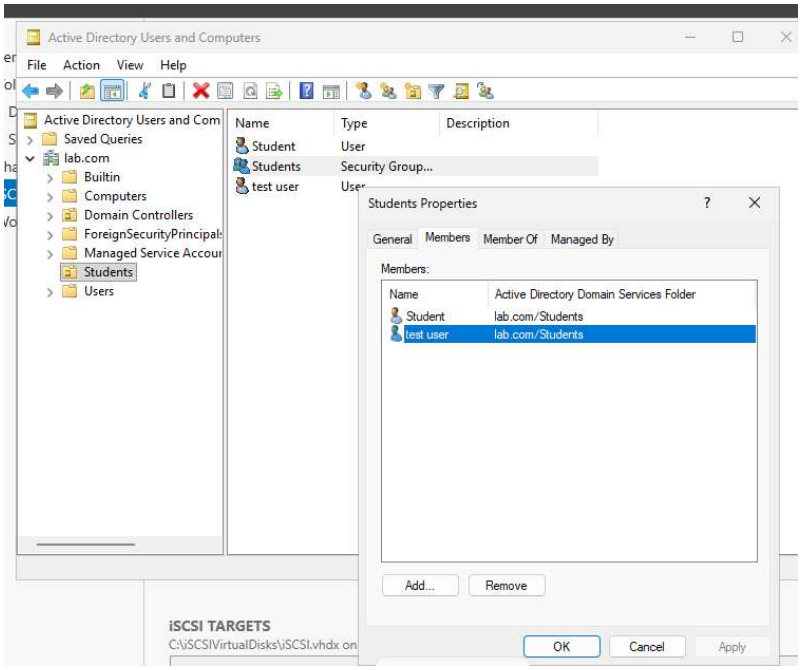


Figure 1: Students security group with members Student and test user.

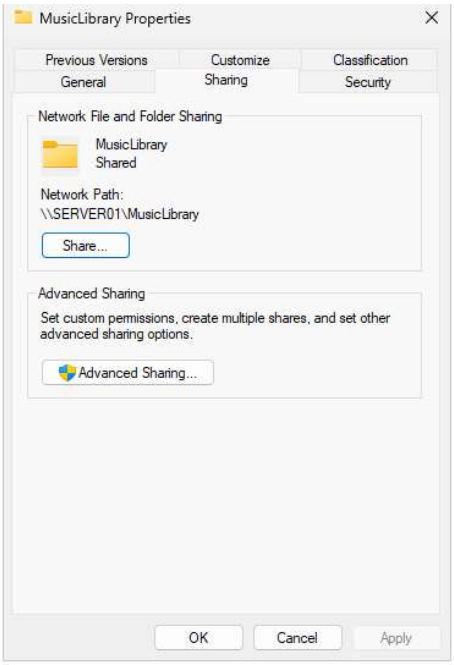


Figure 2: MusicLibrary shared at \\SERVER01\MusicLibrary.

Task 2: Configure NTFS permissions on the 'MusicLibrary' folder so that members of the 'Students' group can read and write files, but cannot delete any files. Hint: Use the Advanced Security Settings to deny delete permissions while allowing modify.

NTFS permissions were set in Properties > Security > Advanced. Inheritance was disabled so that only the entries below apply.

Principal	Type	Access	Applies to
Administrators	Allow	Full control	This folder, subfolders and files
Students	Allow	Read, write & execute (no Delete, no Change permissions, no Take ownership)	This folder, subfolders and files
Students	Deny	Delete, Delete subfolders and files	This folder, subfolders and files

The Allow entry lets Students list, read, create and edit files. The separate Deny entry blocks Delete and Delete subfolders and files. A Deny entry always overrides an Allow, so Students can read and write but cannot delete, even if another group they belong to grants delete. Only Delete permissions were denied, so Students keep full read and write ability.

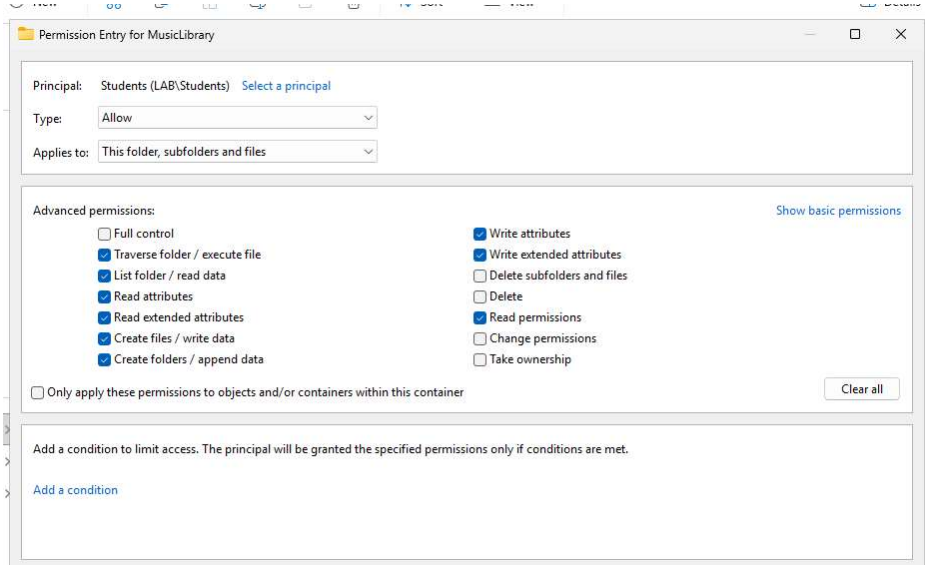


Figure 3: NTFS Allow entry for Students on MusicLibrary (read, write and execute; Delete is not ticked).

Task 3: From a Windows client system, map the 'MusicLibrary' shared folder as a network drive and try to copy, edit, and delete a file. Document which actions succeed or fail, and explain why based on the permissions you set.

On client01, the share was mapped as drive Z: using File Explorer > Map network drive > \ \SERVER01\MusicLibrary, signed in as a member of the Students group. The share permission was Read and the NTFS permissions were as in Task 2.

Action	Result	Explanation
Map \ \SERVER01\MusicLibrary as Z:	Succeeded	Students has access at both the share and NTFS level.
Open / play a file	Succeeded	Read is allowed by both the share and NTFS permissions. Track 2 played in Windows Media Player.
Copy a file into the folder	Failed (Destination Folder Access Denied)	The share permission is Read only. Over the network the stricter of the share and NTFS permissions applies, so the share blocks the write even though NTFS allows it.
Edit a file and save	Failed (Notepad: Access is denied)	Same reason as copy: the share permission is Read only, so saving changes is blocked.
Rename a file	Failed (File Access Denied)	Renaming requires the Delete permission, which Students do not have on NTFS.
Delete a file	Failed (File Access Denied)	Students have no Delete permission, and the NTFS Deny entry blocks it explicitly.

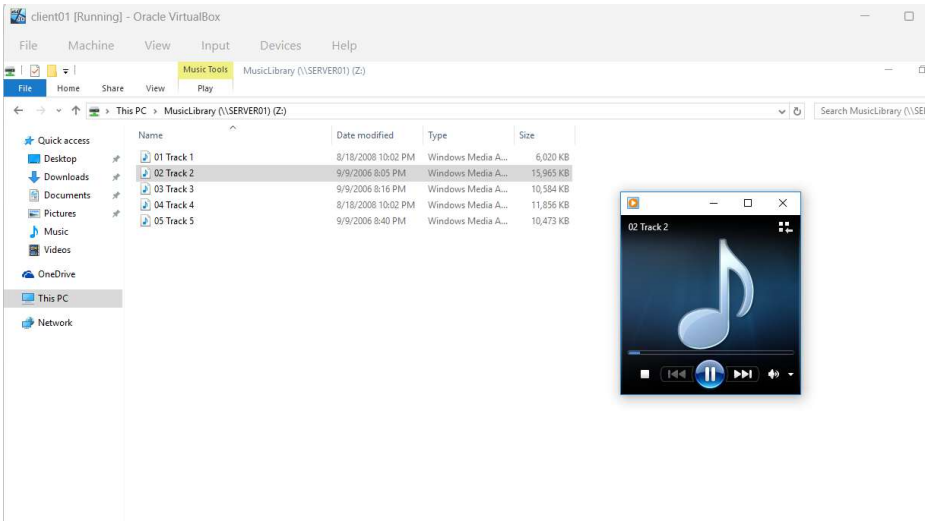


Figure 4: Open and play: Z: mapped to \\SERVER01\MusicLibrary, tracks listed and playing.

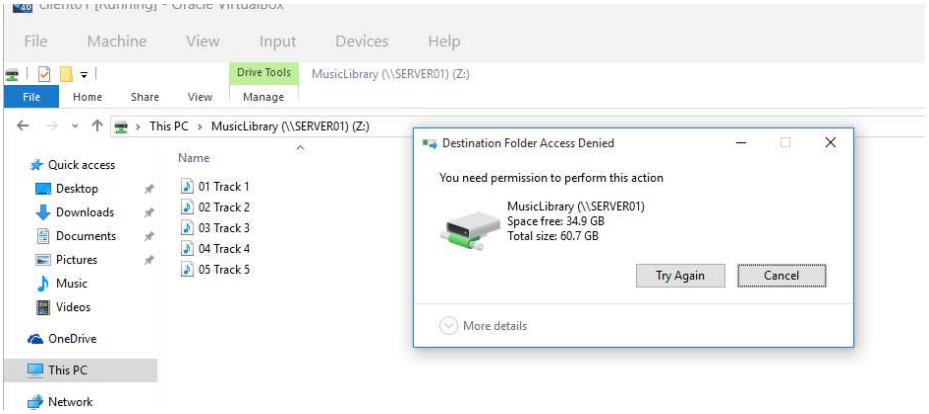


Figure 5: Copy a file in: Destination Folder Access Denied.

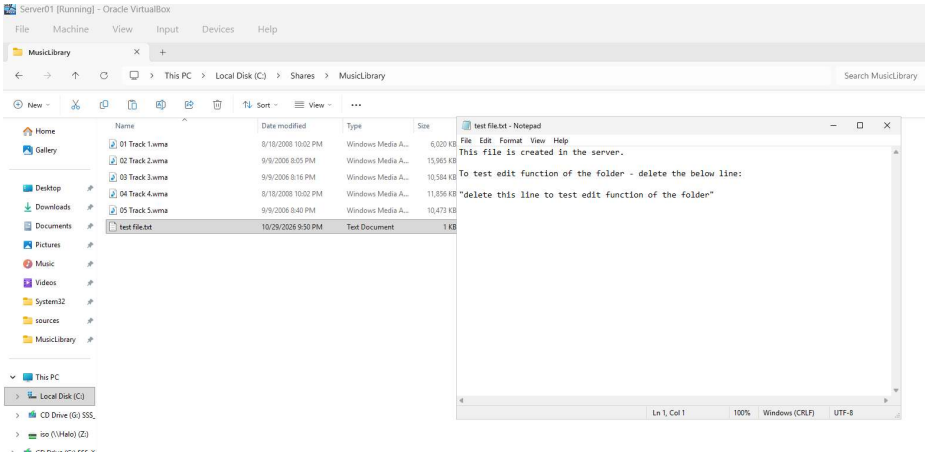


Figure 6: Edit test file created on the server in C:\Shares\MusicLibrary.

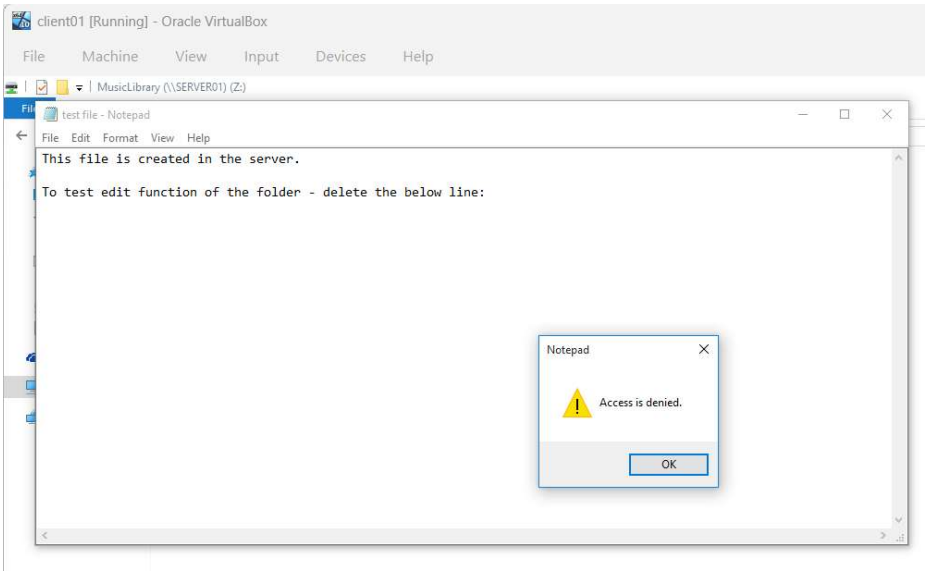


Figure 7: Edit and save from the client: Notepad reports Access is denied.

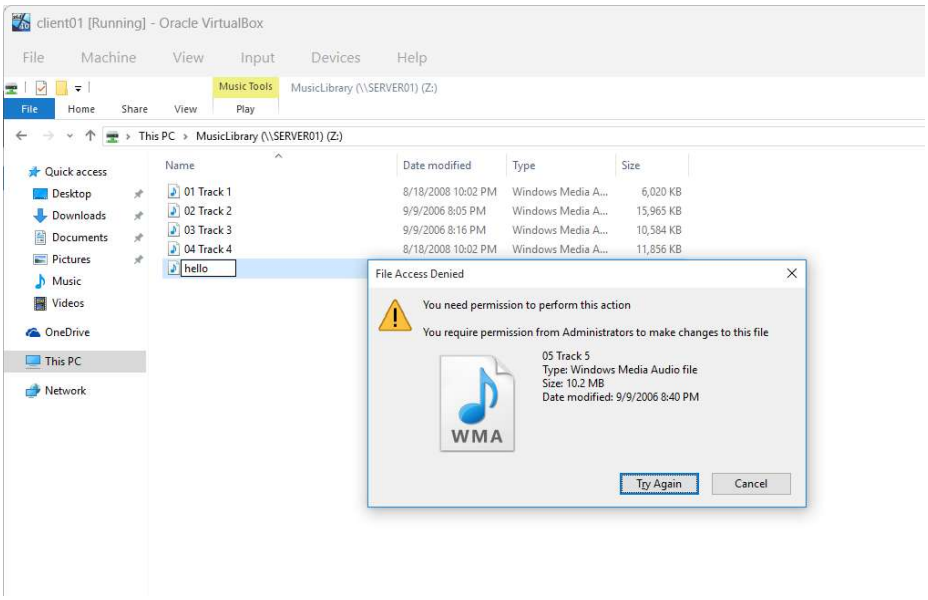


Figure 8: Rename a file: File Access Denied.

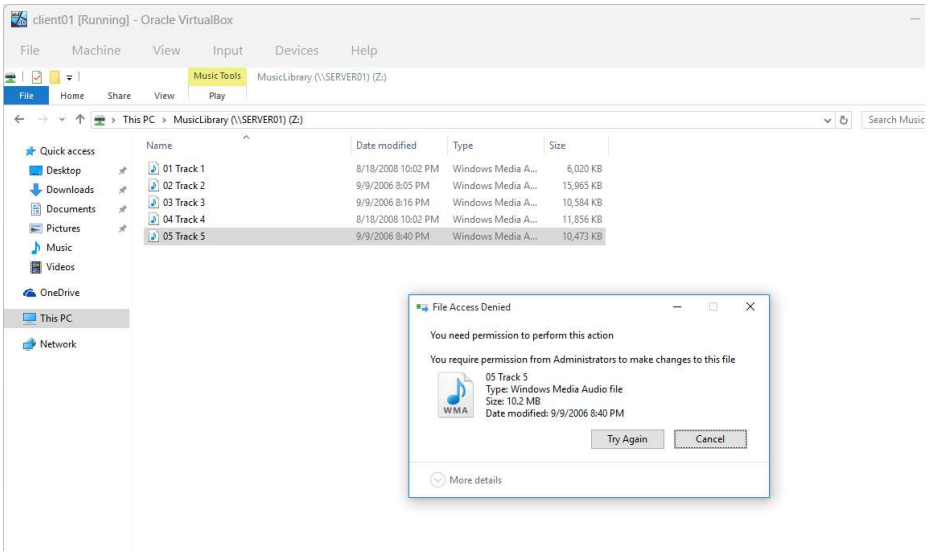


Figure 9: Delete a file: File Access Denied.

The shared folder is protected by two layers, the share permission and the NTFS permission, and a user gets the stricter of the two. Here the share layer (Read) limited Students to reading, and the NTFS layer (no Delete, plus the Deny entry) ensured that files could not be deleted or renamed.

Task 4: Create a second shared folder called 'PhotoShare' and set up share and NTFS permissions so that only your user account has full control. Attempt to access this share from another user account and note the result.

The PhotoShare folder was created and locked down so that only the administrator account used on the server has access.

1. Created the folder C:\Shares\PhotoShare and shared it as PhotoShare (\\SERVER01\PhotoShare).
2. Share permissions: removed Everyone and added the administrator user account (LAB\Administrator) with Full Control (Allow).
3. NTFS permissions: disabled inheritance and converted the inherited permissions into explicit ones, then removed the Users and Authenticated Users entries.
4. Added the same user account with Full control, applied to This folder, subfolders and files.
5. On client01, signed in as a different user and tried to open \\server01\photoshare.

Result: access failed with the message "Windows cannot access \\server01\photoshare. You do not have permission to access \\server01\photoshare."

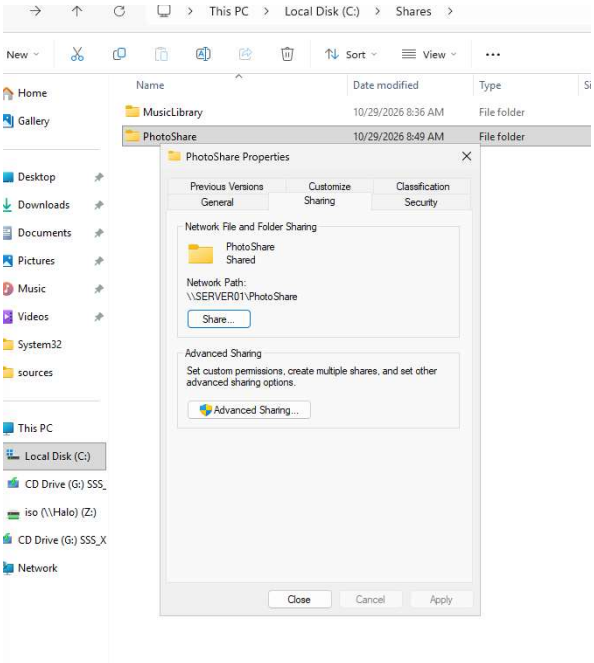


Figure 10: PhotoShare shared at \\SERVER01\PhotoShare.

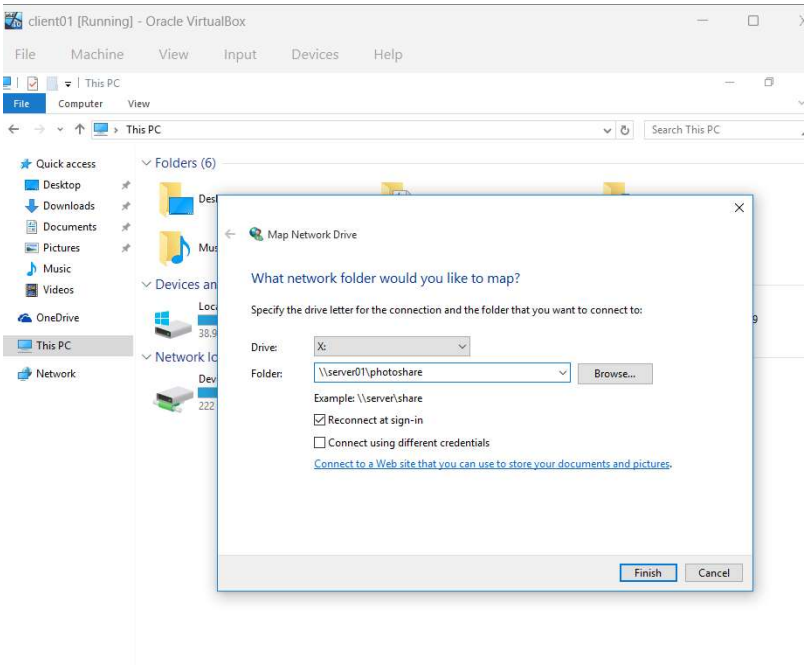


Figure 11: Mapping \\server01\photoshare from client01 as a different user.

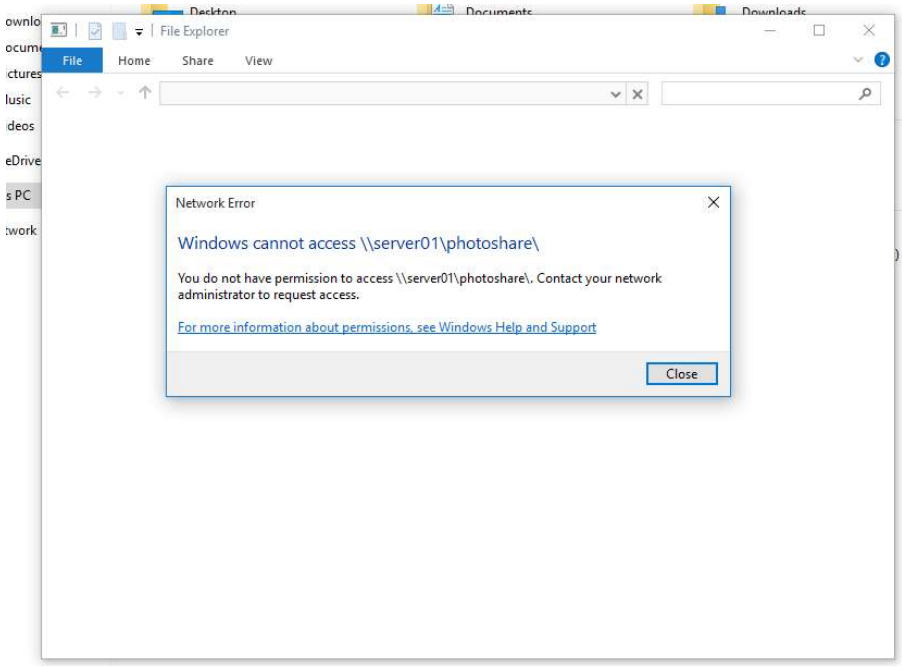


Figure 12: Access to PhotoShare denied for the other user.

Since neither the share permissions nor the NTFS permissions list the other user or any group they belong to. Without an Allow entry, access is denied by default, so the user is stopped at the share before the folder is reached. Signing in as the owner account on the server confirms the share itself works and that the block is caused by permissions only.